Sabia Richards

New York, New York
+1 (469) 268-1161 • sabia.richards@gmail.com

Software engineer with 10+ years of experience and a proven track record of leading teams to create flexible solutions to complex problems. Bridges between advancing research and applying it to create testable outputs that efficiently evaluate performance.

Professional Experience

Staff Machine Learning Engineer
AB InBev, July 2022 – September 2025
Technologies: Python, Tensorflow, C++23, SIMD
- Provided model speedups and memory usage reductions in excess of 100x.
- Led teams handling every part of our technology stack and designing new tools with business feedback.
- Embedded mathematical guarantees of business principles in model architecture.
- Built a distributed framework to optimize marketing budgets across millions of axes with constraints and low latency.
- Helped develop image stitching and recognition software for use by field technicians.

Mobile CI/CD Engineer
ADT Security, August 2021 – April 2022
Technologies: Jenkins, Fastlane, iOS, Android, Gradle, XCode, AppCenter
- Updated and maintained a Jenkins instance
- Created and debugged Jenkins pipelines for a variety of projects for mobile devices.
- Helped design automated builds and releases gated by the Jenkins pipelines.
- Monitored error reports from app use in the wild.

Software Developer
Zemoso, December 2018 – December 2020
Technologies: Java, Swing, Python, Javascript, JMeter, Kubernetes, React-Native, AWS, Android
- Migrated and refactored an Android app to MVVM Architecture.
- Worked on a React Native cross-platform app and native bindings to third party integrations.
- Tested, analyzed, and designed improvements to code and infrastructure for metrics to surpass business requirements.
- Managed, monitored and tested production deployments using the Elastic Kubernetes service on AWS.

Lead Software Developer
smartObjx, August 2017 – Dec 2018
Technologies: C#, Angular, VSTS, SQL, RavenDB
- Built and wrote a system for safe database migrations.
- Developed REST API microservices.
- Automated library generation to allow support for several languages with little to no long term overhead.

Mobile SDK Developer
Apptimize, December 2015 – July 2017
Technologies: A-B-Testing, Android, Automation, iOS, Java, Linux, Objective-C, Jenkins
- Built and deployed new features, third party integrations, and bug fixes for Apptimize's A/B and feature flag testing framework.
- Wrote end to end tests for continuous integration and associated testing procedures.

Speech Engineer

Bumpers, April 2015 – August 2015

Technologies: Android, C++11, Cross-Platform, iOS, Linux, Speech-Recognition

- Explored the possibility of using speech diatrization in a cross-platform mobile app using the SHoUT library.
- Built a library using `kiss_fft` to separate out blocks of speech.
- Evaluated third party libraries on basis of required dependencies and licensing requirements.

Mobile Developer

Fill My Blank, August 2014 – February 2015

Technologies: Android, C++, Cross-Platform, iOS

- Developed and prepared to deploy a mobile application across multiple platforms.
- Focused on optimizing network traffic and on-device code.

Open Source Work

Magic the Gathering: Machine Learning (MtgML)

https://github.com/CubeArtisan/mtgml

Technologies: Python, Tensorflow, Transformers, NLP, Recommendation

- Created several models for predicting the next pick that a player would make in a given Magic the Gathering draft in an arbitrary environment.
- Built models for recommendations on additions and removals to cubes (collections of cards used for drafts).
- Tested approaches for model to build decks given the cards to choose from, had some promising results, but could use more research.

DraftbotOptimization

https://github.com/ruler501/DraftbotOptimization

Technologies: C++, CUDA, SYCL, CMA-ES, Black-Box-Optimization

- The first iteration of what would become the MtgML models using black box optimization techniques.
- Used heterogeneous compute libraries to allow accelerating optimization on whatever compute hardware was available.

CubeArtisan

https://github.com/CubeArtisan/cubeartisan

Technologies: Typescript, NodeJS, React, SolidJS, WebSockets

- A website made for managing Magic the Gathering cubes (collections of cards used for drafting).
- Provided a fully-fledged domain specific filtering language.
- Using the filtering language allowed for creating complicated assignments of cards to packs while maintaining randomness.
- Allowed calculating custom analytics on the complicated random distributions of cards into packs.
- Supported real time and asynchronous multiplayer drafting.

CubeArtisan Infrastructure

https://github.com/CubeArtisan/cubeartisan-infrastructure

Technologies: Kubernetes, CI/CD

- Deployment scripts and configurations for provisioning a kubernetes cluster for CubeArtisan.
- Supported multiple different environments for production and staging.
- Managed deployment of the MtgML models to be consumed by CubeArtisan and other sites.

Mathematics Research Programs

https://github.com/ruler501/Rationals

Technologies: C++11, HPC, Math, Multithreading

- A collection of programs accomplishing a variety of tasks for a mathematics research group.
- Highly optimized and designed to run in parallel on a high performance computing cluster.
- Able to utilize heterogenous compute hardware.

Probability Simulator

https://github.com/ruler501/ProbSim

Technologies: Probability, Statistics, C++17, Functional-Programming, Templates, Variadic-Function

- Developed a library for calculating discrete distributions and transformations on them.
- Supported both a lossless and mildly lossy mode that was much faster.
- Took functions applied to the outputs of a distribution to create new distributions.

Retrieve Green Autonomous Robot Software

https://github.com/ruler501/retgreen

Technologies: C++, OpenCV, PID-Controller, Robotics

- Wrote code to run on a robot designed by my team.
- Searched for and pick up poms of a specified color while avoiding approaching those of other colors.
- Utilized a PID system and custom real time computer vision to determine exact positions.

Corsair Keyboard and Mouse Drivers

https://github.com/ckb-next/ckb-next

Technologies: C, KissFFT, Linux, Drivers

- Built in new visual effects for the Corsair K65/K70/K95/Strafe Driver for Linux and OSX.
- Visualized music using a discrete fourier transform.
- Tracked a heatmap based on what keys were being used.